3. Data Flow & System Behavior

This section presents various diagrams that describe how data flows through the system and how different components interact within the Restaurant Finder application. These diagrams provide a clear understanding of system processes, workflows, and structure.

3.1 Data Flow Diagrams (DFD)

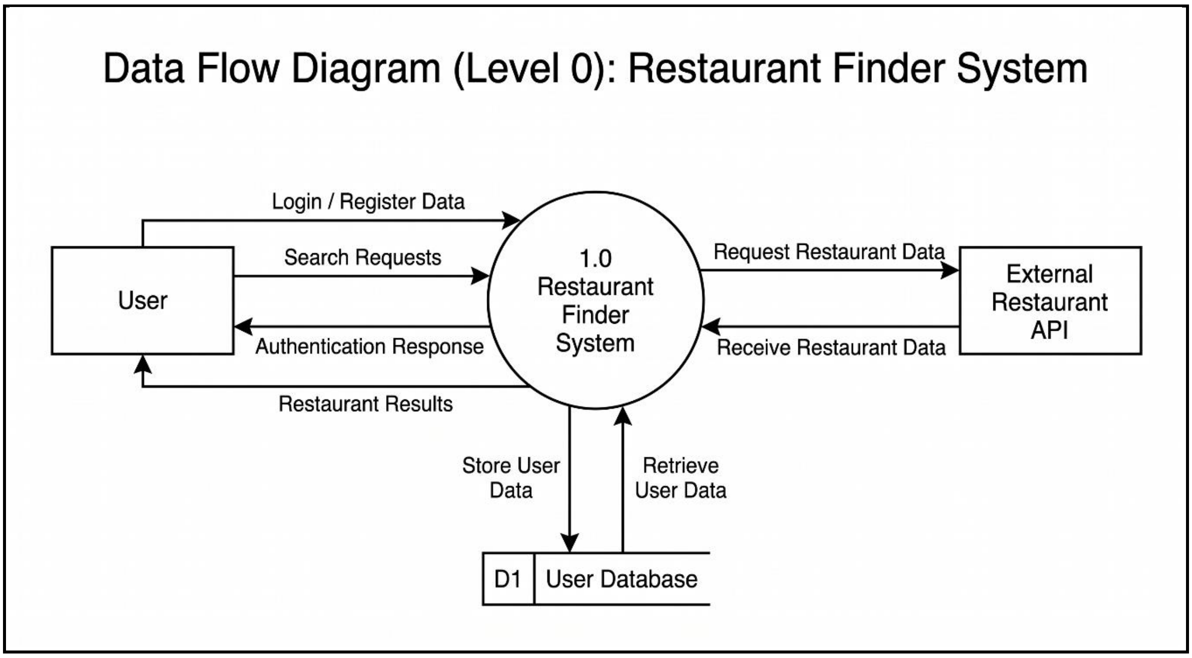
- Context-Level DFD: This diagram provides a high-level overview of the system as a single process interacting with external entities and databases.

- In this diagram, the entire system is represented as a single process called "Restaurant Finder System."
  The User interacts with the system by sending inputs such as login/register data or search requests.

- The system then communicates with the External Restaurant API to fetch restaurant data and returns the results back to the user.

- It also interacts with the User Database to store and retrieve user information when needed, such as during authentication.

- This diagram provides a high-level overview of the system without showing internal details.



Level 1 DFD: This diagram breaks down the system into core processes such as authentication, search, details, and favorites.

- In this diagram, the system is broken down into smaller processes to show how it works internally.

- The first process is User Authentication, which handles login and registration. It interacts with the database to store and retrieve user data.

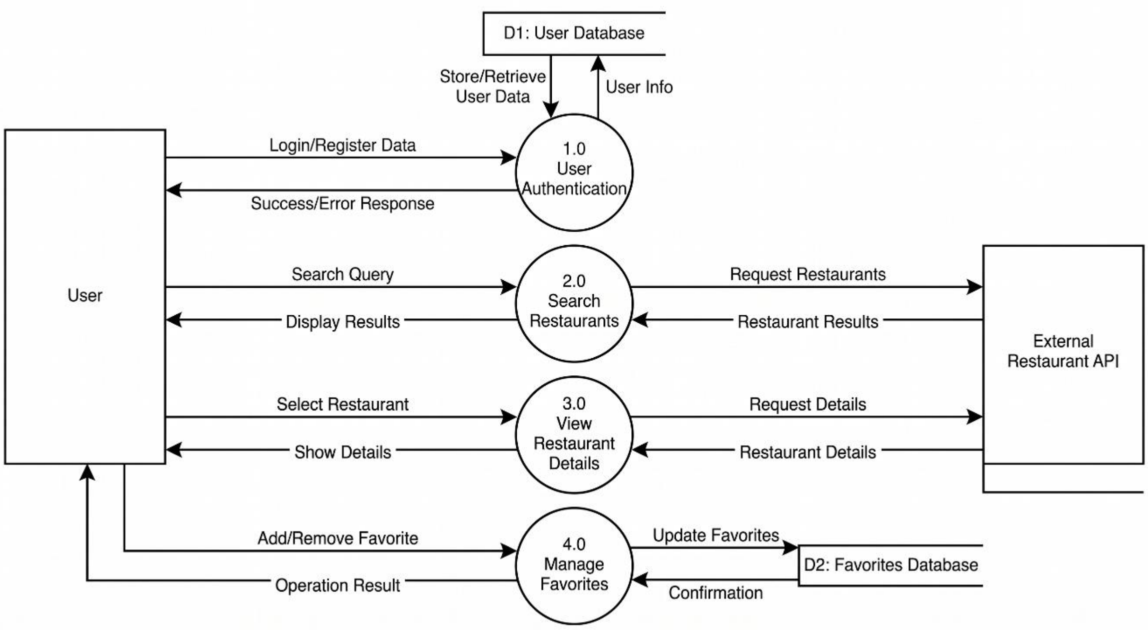
- The second process is Search Restaurants, where the user sends a search query, and the system requests data from the API and returns a list of restaurants.

- The third process is View Restaurant Details, where the system fetches detailed information about a selected restaurant from the API and displays it to the user.

- The fourth process is Manage Favorites, which allows the user to add or remove restaurants from their favorites list, and this data is stored in the database.

- This diagram clearly shows how data flows between the user, system processes, database, and external API, giving a detailed understanding of the system behavior.

Figure 2: Level 1 DFD



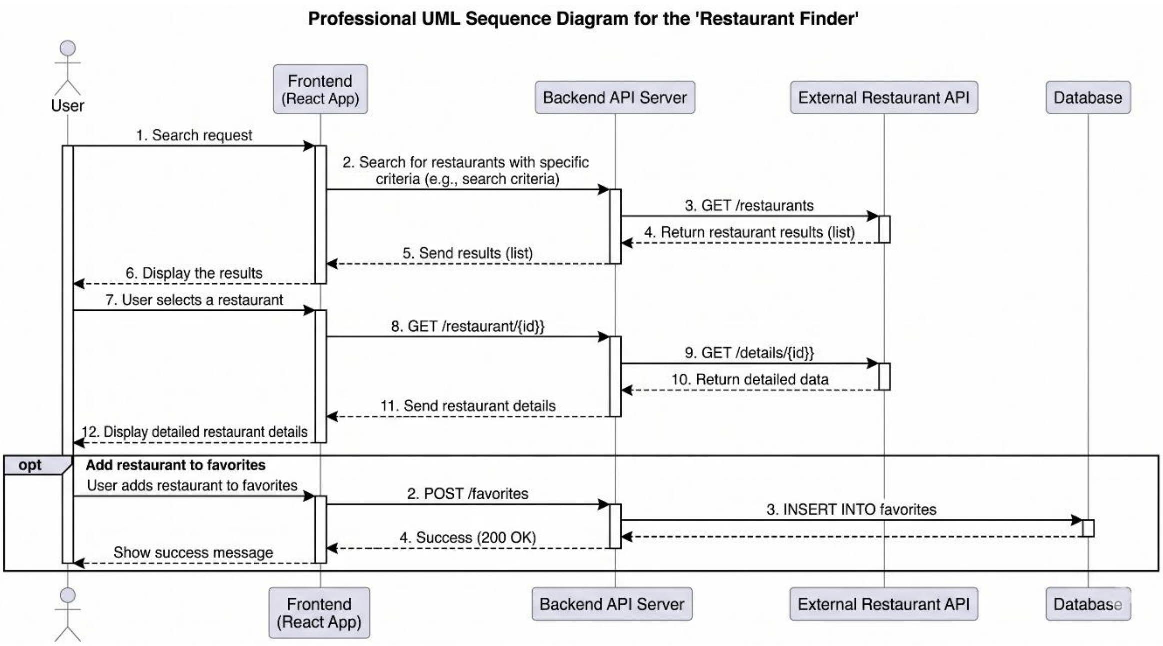
3.2 Sequence Diagram

The Sequence Diagram represents the interaction between system components over time. It illustrates how requests and responses are exchanged between the different parts of the system, including the user/admin, frontend application, backend API, database, and external services.

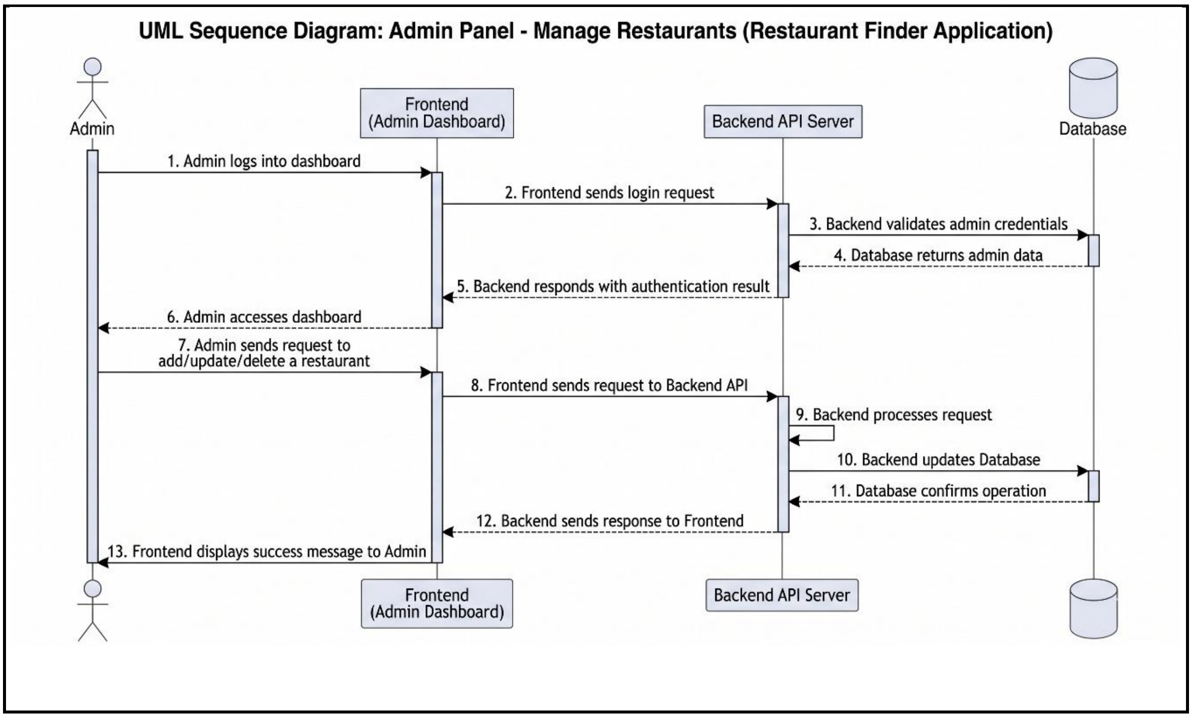
In this project, two main sequence diagrams were designed to describe the system behavior:

- **User Sequence Diagram:**
  This diagram demonstrates the flow of interactions when a user accesses the system. It includes browsing restaurants, viewing restaurant details, and performing actions such as making a reservation. The diagram shows how the frontend communicates with the backend API, which in turn interacts with the database and external services when required.

Figure 3: Sequence Diagram

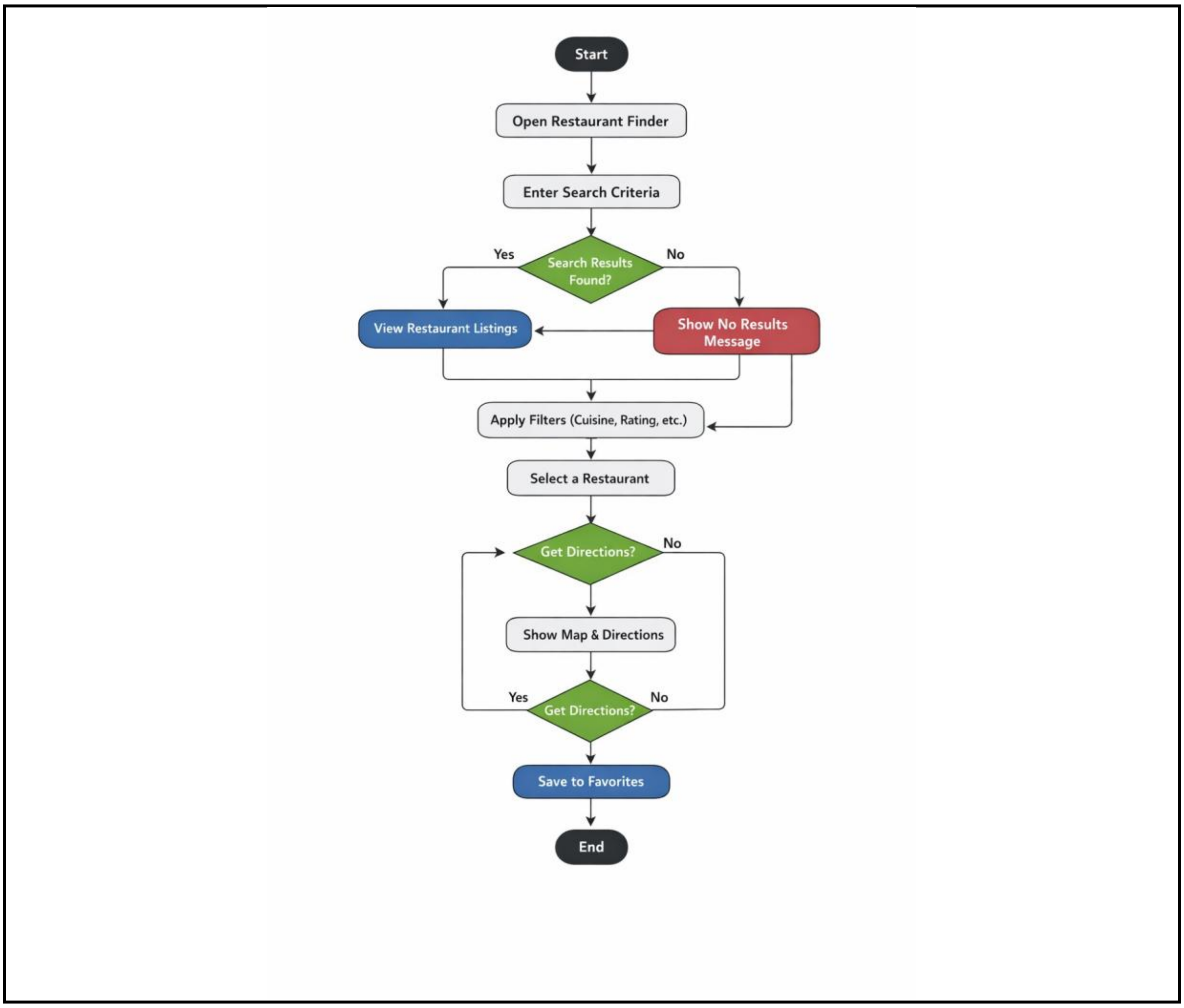


- **Admin Sequence Diagram:**
  This diagram illustrates the interactions of the administrator with the system. It covers administrative operations such as logging into the dashboard, adding new restaurants, updating or deleting existing data, and managing user orders or reservations. It also highlights the authentication process and secure communication between the admin dashboard, backend API, and database.



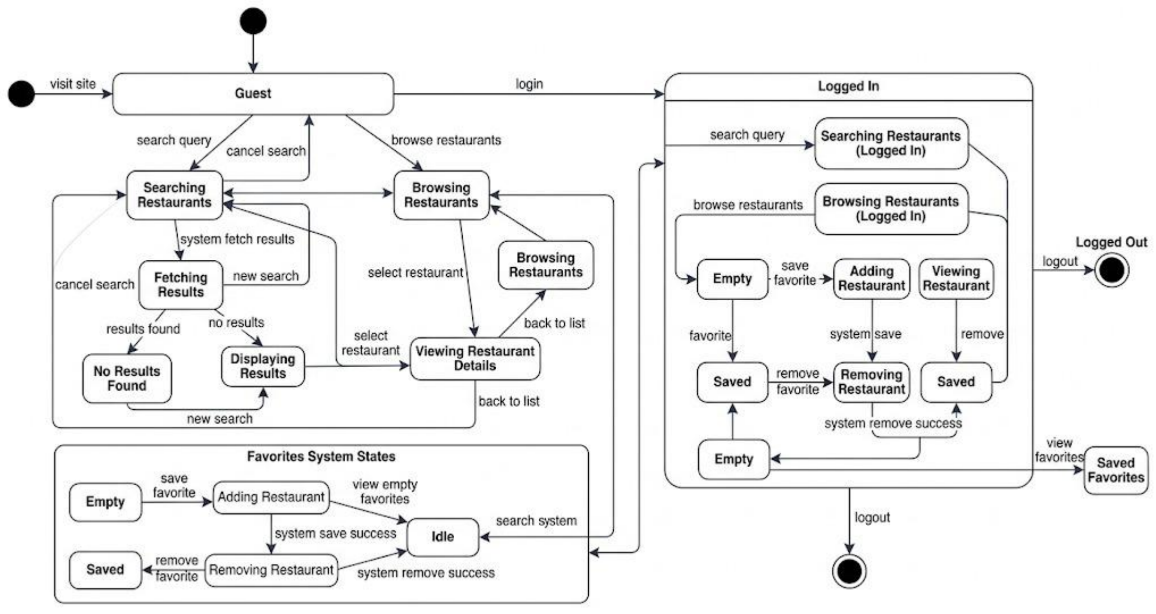
3.3 Activity Diagram

The Activity Diagram visualizes the workflow of user actions and system processes. It describes the sequence of activities from user interaction to system response.



3.4 State Diagram

The State Diagram represents different states of an object and how it transitions between them based on user actions or system events.



3.5 Class Diagram

The Class Diagram defines the system structure by showing classes, attributes, methods, and relationships between different components in the system.

Figure 6: Class Diagram